

Introduction to Mathematics and Optimization

January 2024

PROBLEM 1

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function

$$f(x, y) = x^2 - y$$

and $C \subseteq \mathbb{R}^2$ the subset

$$C = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1, (x + 1)^2 + (y - 1)^2 \leq 1\}.$$

We consider the optimization problem

$$\begin{aligned} \min f(x, y) \\ (x, y) \in C \end{aligned} \tag{1}$$

- (a) Prove that f is a convex function. Is f a strictly convex function?
- (b) Show that C is a subset of $\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}$ and that C is a bounded subset of $\mathbb{R}^2$.
- (c) Show that C is a compact and convex subset of $\mathbb{R}^2$.
- (d) Which result in the notes ensures that the optimization problem (1) can be solved, and why?
- (e) Show from the KKT conditions that $(-1, 0)$ is not an optimal solution to (1).
- (f) How do we know that the KKT conditions corresponding to (1) actually have a solution?

PROBLEM 2

Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be the function given by

$$f(x, y, z) = x^2 + y^2 + z^2 - xy + yz - xz$$

and let $C \subseteq \mathbb{R}^3$ be the subset given by

$$C = \{ (x, y, z) \in \mathbb{R}^3 \mid x \geq 0, y \geq 1, z \geq 0 \}.$$

- (a) Show that f is a strictly convex function.
- (b) Show that C is not a bounded subset of $\mathbb{R}^3$.
- (c) Write down the KKT conditions for the optimization problem

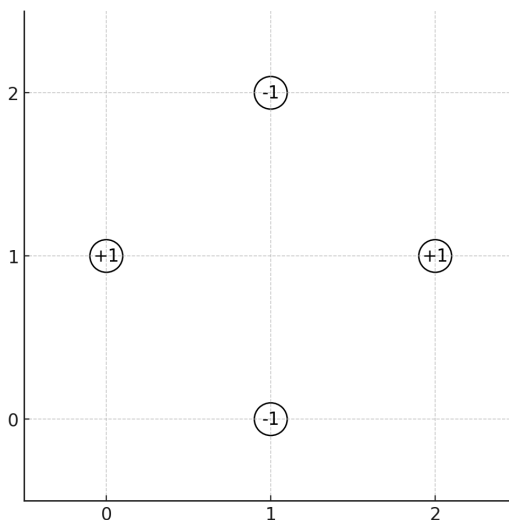
$$\begin{aligned} \min f(x, y, z) \\ (x, y, z) \in C, \end{aligned} \tag{2}$$

and show that $(\frac{1}{2}, 1, 0)$ is an optimal solution to (2) and that the KKT conditions have a unique solution.

- (d) Is $f(x, y, z) \geq 0$ for all $x, y, z \in \mathbb{R}$?

PROBLEM 3

- (a) The points $(0, 1), (1, 0), (1, 2), (2, 1) \in \mathbb{R}^2$ are assigned labels $+1, -1, -1, +1$ respectively:

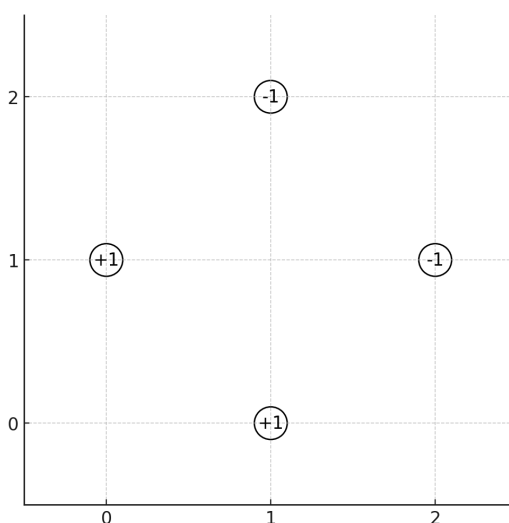


Show that if the line $L = \{(x, y) \in \mathbb{R}^2 \mid ax + by + c = 0\}$ separates the points, then

$$\begin{aligned} b + c &> 0 \\ 2a + b + c &> 0 \\ a + c &< 0 \\ a + 2b + c &< 0. \end{aligned} \tag{3}$$

- (b) Show using Fourier-Motzkin elimination that the inequalities in (3) cannot be solved, and hence that there exists no line that separates the points.

- (c) We now assign the same points $(0, 1), (1, 0), (1, 2), (2, 1)$ labels $+1, +1, -1, -1$ respectively:



Set up an optimization problem for finding the best separating line $L = \{(x, y) \in \mathbb{R}^2 \mid ax + by + c = 0\}$ that now separates the points. Show, without drawing (without geometric considerations), that $a = -1, b = -1$, and $c = 2$ is an optimal solution to the optimization problem.